

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Commercial APIs expose model names, not proof. If cheaper inference can masquerade as the advertised model, integrity becomes an economic security problem.

9

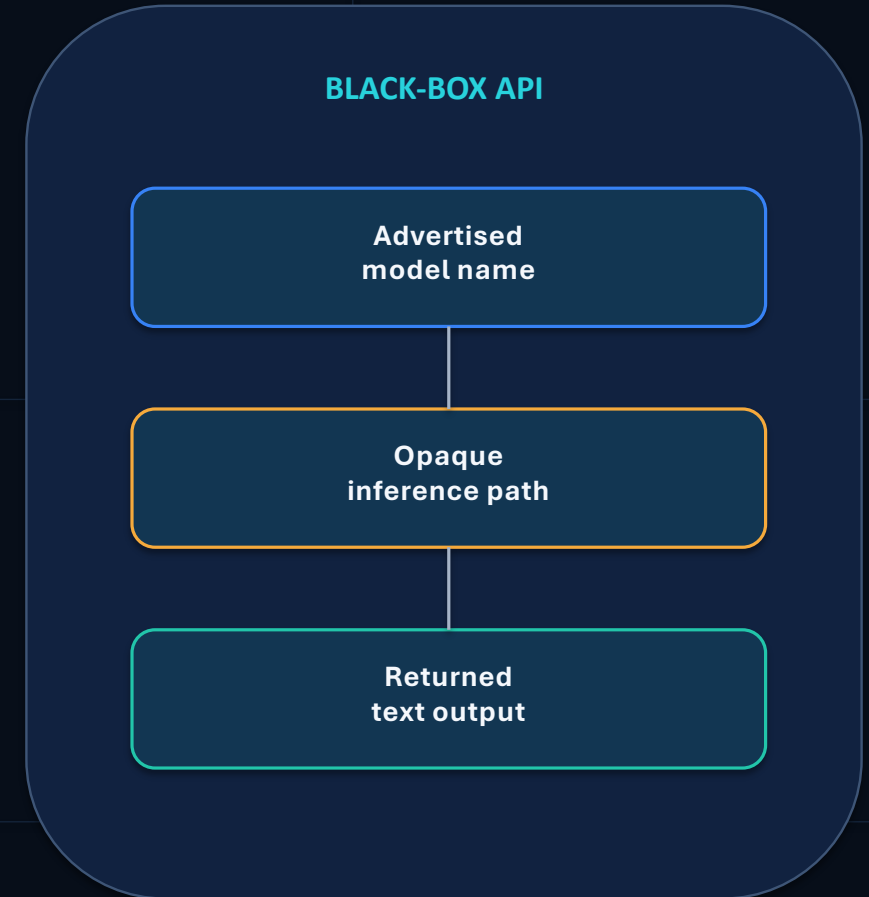
slide evidence chain

3

software failure modes

1

actionable assurance path



The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Given API samples, can an auditor distinguish the specified distribution from the actual served distribution?

$H_0: y \sim P_{\text{spec}}(y \mid x)$ vs. $H_1: y \sim P_{\text{actual}}(y \mid x)$

Economic motive: reduce GPU costs while maintaining advertised pricing.

Auditor sees only prompts and outputs

Provider controls routing, runtime, weights, and metadata exposure.



Smaller model

8B instead of 70B



Quantized variant

FP8 / INT8 instead of FP16



Fine-tuned update

silent model version drift



Different family

architecture swap

Cost

Trust

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Original vs. FP8 scores across four benchmarks for five model families. Many FP8 differences are small relative to reported uncertainty, especially on MMLU and GSM8K.

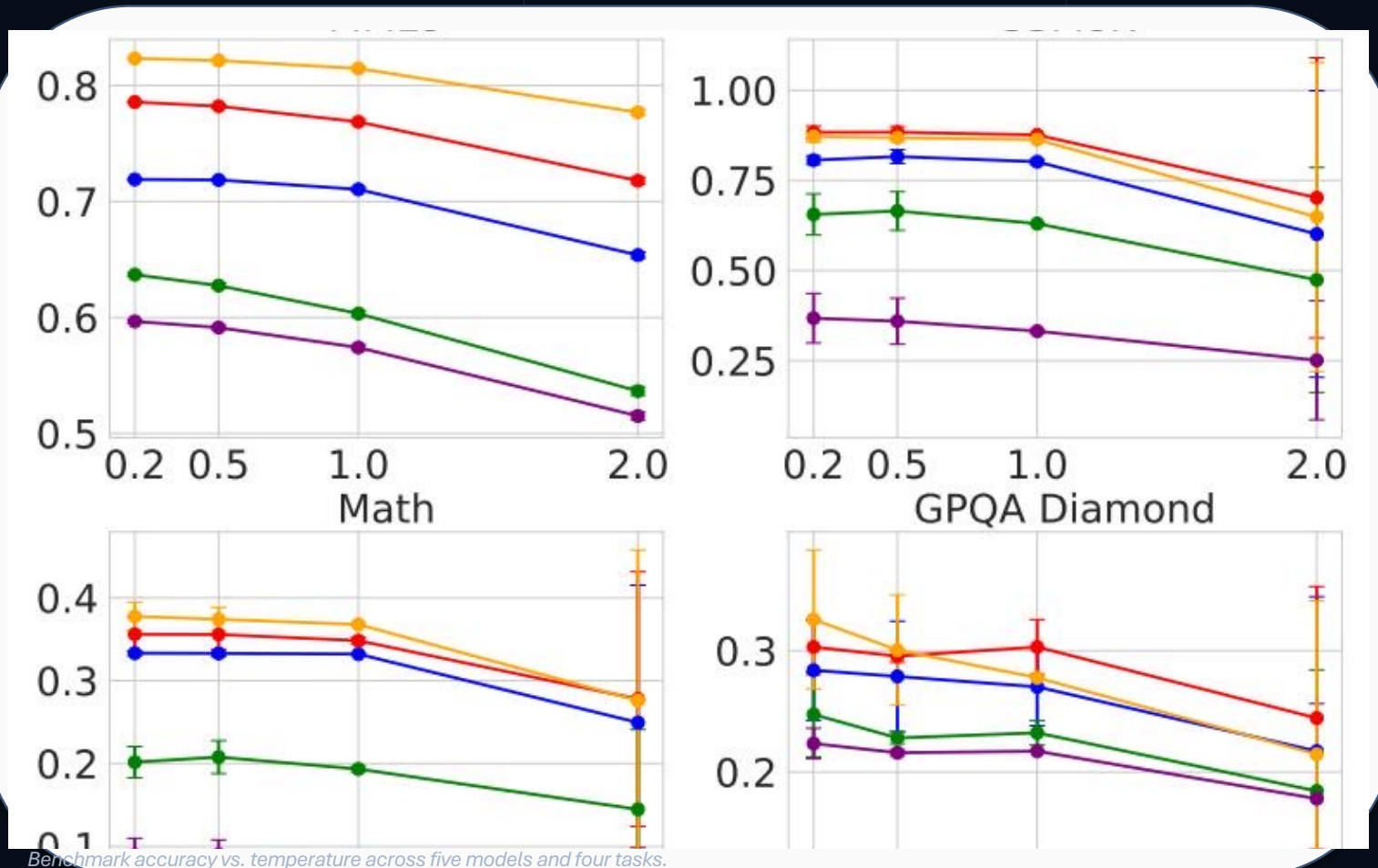
Model family	MMLU Orig / FP8	GSM8K Orig / FP8	MATH Orig / FP8	GPQA Diamond Orig / FP8
Llama-3-8B-Instruct	62.69±0.18 / 62.43±0.26	61.14±3.47 / 60.90±4.10	20.65±3.43 / 14.91±2.49	22.62±0.22 / 20.14±0.24
Llama-3-70B-Instruct	78.05±0.08 / 77.88±0.13	88.06±1.44 / 87.35±1.24	35.69±1.33 / 35.75±1.16	29.60±0.51 / 33.12±0.30
Gemma-2-9b-it	71.86±0.08 / 71.92±0.11	81.80±1.35 / 79.41±1.14	33.41±0.28 / 32.53±0.34	28.64±2.97 / 27.81±3.22
Qwen2-72B-Instruct	82.18±0.08 / 81.98±0.08	86.72±1.00 / 86.82±0.97	37.39±1.41 / 37.67±1.39	29.93±2.71 / 31.08±1.96
Mistral-7B-Instruct-v0.3	59.15±0.10 / 58.77±0.13	35.90±4.54 / 32.20±4.02	8.94±1.24 / 7.68±1.23	21.60±0.17 / 22.72±0.19

Signal is subtle: FP8 often remains within error bars.

Benchmark accuracy alone does not certify model identity.

Small drops make covert quantization attractive.

Higher Temperature Amplifies Performance Gaps but Increases Noise



Interpretation

- Temperatures evaluated: 0.2, 0.5, 1.0, 2.0.
- Higher temperature degrades performance across models.
- Noise grows as sampling moves away from deterministic behavior.
- Substitution gaps narrow or become unstable under noisy decoding.

Consequence: statistical tests must separate model identity from sampling variance.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

	BERT Acc	T5 Acc	GPT2 Acc
ruct-FP8	50.55	50.10	50.45
ruct-INT8	51.60	49.90	51.30
P8	49.95	50.30	51.20
NT8	49.00	49.70	51.65
struct-FP8	50.55	49.75	48.70
struct-INT8	49.70	50.75	50.50
ruct-FP8	50.05	50.25	48.75
ruct-INT8	50.75	50.55	49.45

Same Llama-3-8B model; log-probabilities vary across frameworks and GPU hardware.

Why the fingerprint breaks

transformers
4.40 vs 4.50

vLLM
0.6.2 vs 0.8.2

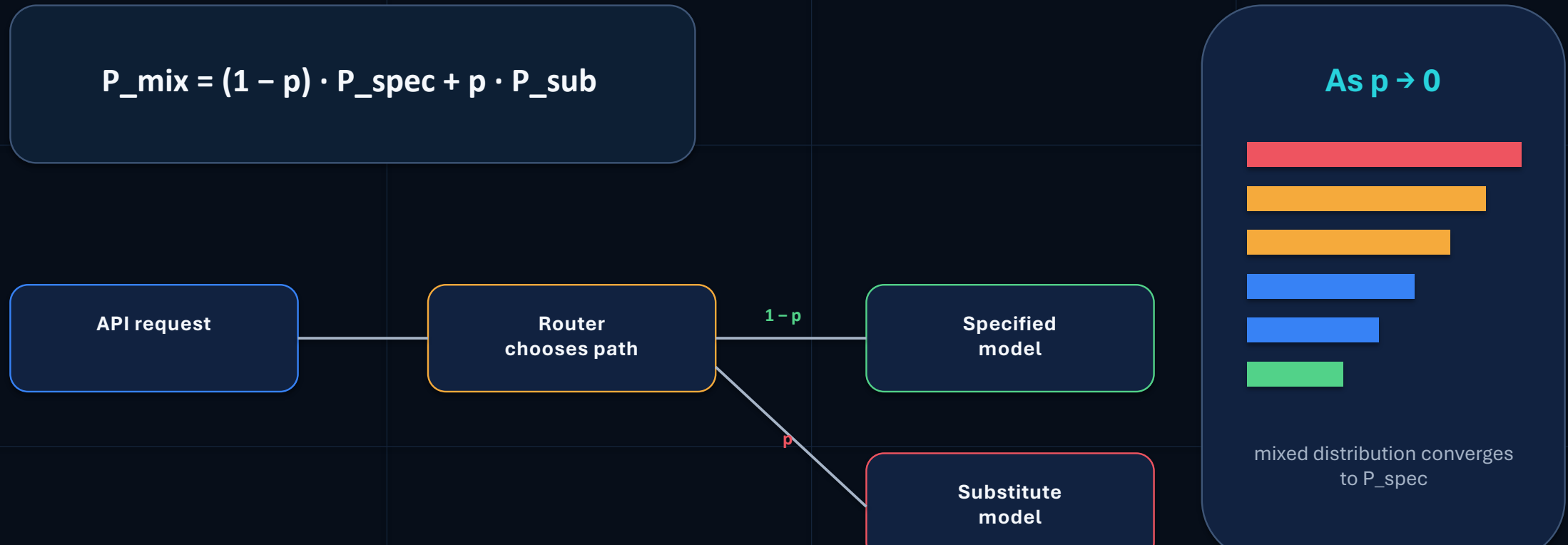
GPU
A100 vs H100

An auditor cannot reliably distinguish legitimate production variation from actual substitution when token-level metadata shifts under normal infrastructure changes.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

A provider need not substitute every query. Mixing cheaper responses with probability p drives the observed distribution toward the specified model as p decreases.

$$P_{\text{mix}} = (1 - p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$$



Practical limitation: subtle or randomized substitutions require impractically large query budgets for text-output statistical tests.

Software-Only Auditing Methods Are Fundamentally Unreliable

The failure is not one technique; it is the measurement surface. Text, metadata, and benchmark prompts remain controlled or confounded in black-box APIs.

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Subtle quantization and randomized routing make output distributions hard to separate.	Impractically large query budgets; fails against low-rate substitution.
Log-probability methods	Production nondeterminism shifts token log-probs across frameworks and hardware.	Cannot separate legitimate infrastructure variation from substitution.
Adversarial countermeasures	Provider can benchmark-detect prompts or route only a fraction p to substitutes.	Audits become brittle, query-prohibitive, and gameable.

Software auditing observes symptoms. Model integrity requires proving the executed weights and code.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEEs shift the audit target from behavioral inference to hardware-backed attestation of code and data execution.



Software-only audit

- Behavioral outputs
- Statistical evidence
- Confounded by sampling/runtime
- Gameable by routing

TEE-backed inference

- Executed code + weights
- Cryptographic attestation
- Robust to output noise
- Practical deployment path

Practical path: NVIDIA confidential computing on H100 GPUs; modest performance overhead.

Key Takeaways: Close the Verification Gap with Hardware Attestation

- 1 Economic pressure creates a credible incentive for covert model substitution.
- 2 FP8 quantization barely affects many benchmark scores, weakening benchmark audits.
- 3 Inference nondeterminism defeats log-probability fingerprinting in production stacks.
- 4 Randomized routing and benchmark evasion undermine software-only detection.
- 5 TEEs provide the actionable route to verifiable inference via hardware attestation.

Bottom line

Do not infer identity from behavior when the provider controls the behavior. Require attestation of the runtime, code, and weights.

Verification shifts from probabilistic auditing to cryptographic evidence.

Recommended next step for API integrity programs: evaluate TEE-backed inference and remote attestation requirements before relying on black-box software tests.